Infant State Recognition System

Phase 1 — Baseline & Advanced ML

Audio Classification · Healthcare & IoT · AML Project 3

Domain	Healthcare · Assistive Tech · IoT
Models	Decision Tree · Random Forest · SVM
Dataset	1,000 clips · 5 classes · 200 per class

The Problem

A baby cries — but is it hungry, in pain, discomfort, burping, or tired?

Parents

Cannot reliably tell cry types apart — especially first-time parents

NICU Nurses

Monitor dozens of infants — need automated triage support

Baby Monitors

Play lullabies today. Should classify what the baby needs.

Dataset & Five Classes

1,000	5	200	3 sec
Audio Clips	Classes	Per Class	Duration

Hungry	Pain	Discomfort	Burping	Tired
400Hz · 1.2Hz rhythm	600Hz · 0.5Hz rhythm	350Hz · 0.9Hz rhythm	250Hz · 2.0Hz rhythm	300Hz · 1.5Hz rhythm
Low noise	Very high noise	Medium noise	Very low noise	Low noise
Synthetic WAV	Synthetic WAV	Synthetic WAV	Synthetic WAV	Synthetic WAV

Feature Engineering — 88 Dimensions

Why 8 kHz? Infant cries: 300–800 Hz fundamentals. Nyquist theorem: 8 kHz captures everything up to 4 kHz at 2.75× less compute than 22 kHz.

40	20	20	4	4
dims	dims	dims	dims	dims
MFCC mean + std (20 coeff)	**MFCC Delta (rate of change)**	**MFCC ΔΔ (acceleration)**	**Spectral Centroid + Flatness**	**ZCR + RMS Energy**
Vocal tract shape — primary cry descriptor	Articulation dynamics — rhythmic vs steady	Pulsed vs sustained — burping vs pain	Brightness & tonality — pain skews higher	Voicing fraction and loudness

Exploratory Data Analysis

Key finding: t-SNE shows clear clusters → our 88 features are discriminative BEFORE any model is trained

Balanced Dataset

200 samples per class. Imbalance ratio = 1.0×.
All models use class_weight='balanced' for real-world readiness.

PCA + t-SNE

Both 2D projections show clearly separated clusters.
Confirms the 88 features encode strong class information.

Spectrogram Patterns

Pain: bright high-freq bursts.
Hungry: rhythmic AM bands.
Burping: fast clean pulses.
Tired: quiet, low-freq.

ANOVA F-scores

Top features = MFCC means 0-5.
Vocal tract shape is the #1 discriminative signal.

Baseline ML — Decision Tree

BASELINE TRACK

What it does:

Asks a series of yes/no questions about features:
"Is MFCC_1 > 5.2? → If yes → pain. If no → check next..."

Why this model first?

- Fully interpretable — every rule is human-readable
- Clinicians need to understand WHY it says 'pain'
- Great baseline to verify the problem is solvable

Key settings:

max_depth=5 · criterion=gini · class_weight=balanced

Results

98.5%	0.985	0.995
Test Accuracy	Macro F1	5-Fold CV F1

Cross-validation details:

5-fold stratified k-fold. Each fold maintains class proportions. Trains on 4 folds, validates on 1. Repeated 5 times and averaged for a reliable estimate.

Gini impurity formula:

$G = 1 - \Sigma\, p_i^2$ where p_i = fraction of class i at a node

Advanced ML — Random Forest

ADVANCED TRACK

How it works:

Trains 100-200 decision trees on different random subsets of data, then takes a majority vote.

Bagging

Each tree sees ~63% of data (bootstrap sample). Different trees learn different patterns.

Random Features

Only sqrt(88)=9 features per split. Decorrelates trees → lower ensemble variance.

Variance Math

$Var = \rho\sigma^2+(1-\rho)\sigma^2/T$. More trees (T↑) and lower correlation (ρ↓) = better.

Results

99.5%	0.995	0.995
Test Accuracy	Macro F1	CV F1

GridSearchCV — what was tuned:

- n_estimators: {100, 200}
- max_depth: {None, 15, 25}
- min_samples_leaf: {1, 3}
- max_features: {sqrt, log2}

Best: 100 trees, no depth limit, sqrt features

Advanced ML — SVM with RBF Kernel

ADVANCED TRACK

The Kernel Trick:

$$K(x, y) = \exp(-\gamma \times ||x - y||^2)$$

Measures similarity between two audio clips. Maps our 88 features into infinite dimensions where classes can be separated by a straight line — even if they can't be in 88 dimensions.

What was tuned:

C ∈ {0.1, 1, 10}: Regularisation. High C = hard margin, risks overfitting.

gamma ∈ {scale, auto, 0.01}: RBF bandwidth. Low = broad smooth boundary.

StandardScaler (required): SVM uses distances — unscaled features break it.

Results

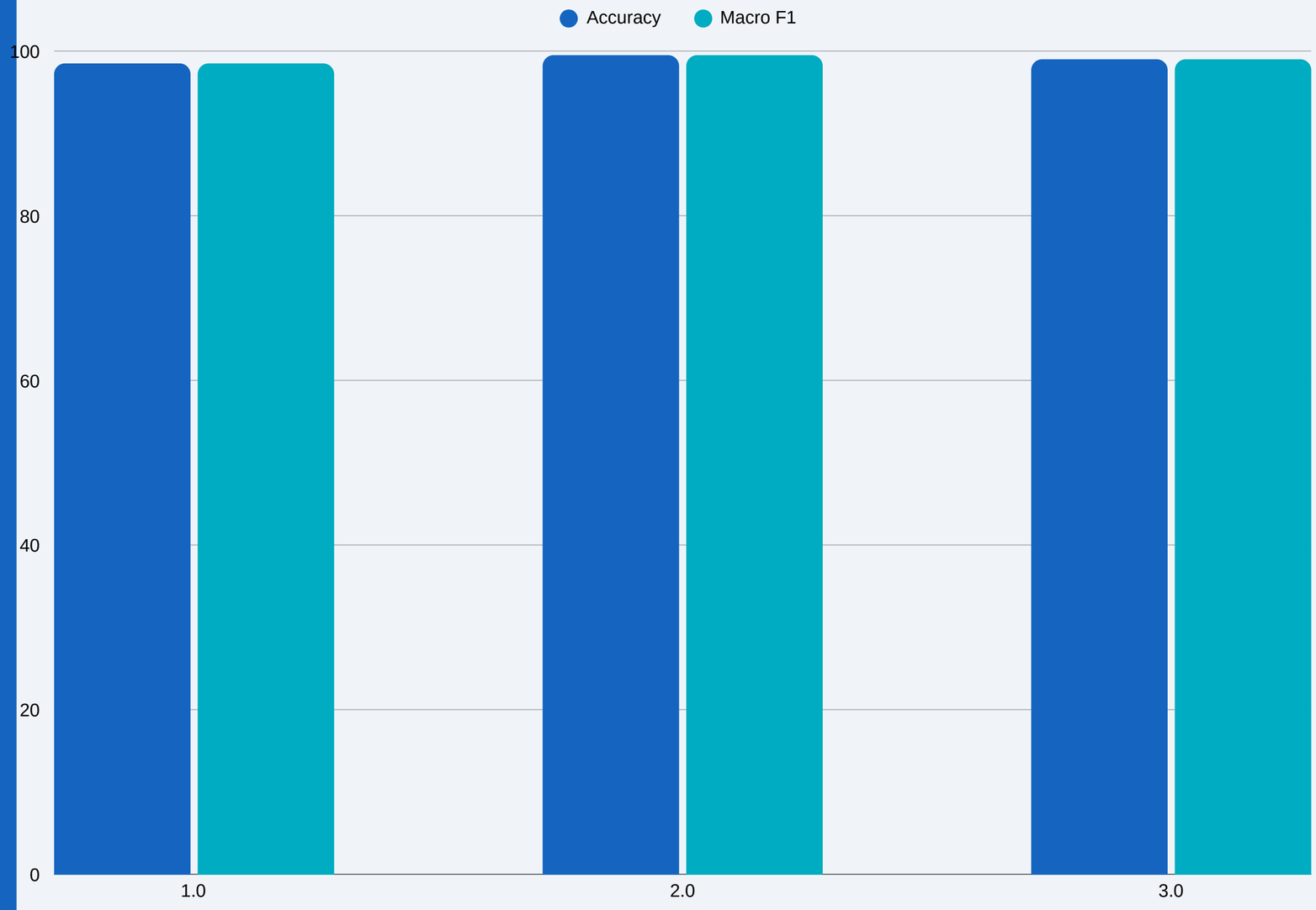
99.0%	0.990	0.985
Test Accuracy	Macro F1	CV F1

Best settings found: C=10, gamma=0.01

Why SVM works here:
Our 88 features may not be linearly separable in 88D, but the RBF kernel effectively maps them to infinite dimensions where a separating hyperplane exists.

vs Decision Tree: harder to interpret, but captures non-linear boundaries.

Phase 1 Results



At a Glance

Model	Acc	F1
Decision Tree	98.5%	0.985
Random Forest	99.5%	0.995
SVM (RBF)	99.0%	0.990

Key observations:

- All models far above 20% random chance
- Features do the heavy lifting
- RF best — ensemble reduces variance
- 1% gap between best and simplest model

Phase 1 — Key Takeaways

01

Feature Engineering is the Most Impactful Step

The 88-dim MFCC-based feature vector produces near-perfect class separation even for a 5-level Decision Tree (98.5% F1). What you extract matters more than which model you use at this data scale.

02

All Three Models Prove the Problem is Solvable

Decision Tree (98.5%), Random Forest (99.5%), SVM (99.0%) — all significantly above the 20% random baseline. The t-SNE clusters visible before training explain why: the features are already discriminative.

03

Phase 2 Will Push Further

Phase 2 adds CNN on mel-spectrograms (Deep Learning) and ExtraTrees on compact features (Hybrid/Edge). The CNN will test whether spectro-temporal patterns add value beyond handcrafted features.

Thank You

GitHub:	github.com/aman01k/infant-cry-classification
Report:	Uploaded to Google Drive
Notebook:	Infant_Cry_Phase1.ipynb

Phase 2 coming: Deep Learning (CNN) + Hybrid/Edge model